RRA

Lab9 (Parteneriat cu laburile scrise de PETrascu)

O mica reamintire:

Fie $G = \Phi^{\sim} J \Phi$, unde $\Phi^{\sim}, \Phi \in \mathscr{RH}_{\infty}$. Asta este o factorizare $J-\mathrm{spectrala}$, cu Φ transformare de congruenta si

$$J = \begin{bmatrix} -I_{\mathrm{m}1} & 0 \\ 0 & I_{\mathrm{m}2} \end{bmatrix}$$

De asemenea, Youla ce facea ete sa ne dea clasa tuturor compensatoarelor care respecta anumite cerinte.

Acum vine intrebarea: cum se combina astea? Prin

Factorizari inner-outer

Sa vedem cum arata astea:

$G(s) = G_i(s) G_0(s)$, unde G_i este factorilzare inner $\in \mathscr{RH}_{\infty}$ si cu inversa stabila a.i $G_i^{\sim} G_i = I$. G_o este factorizare outer. Aici $\exists G_o^{\#} \in \mathscr{RH}_{\infty}$ unde $G_o(s) G_o^{\#}(s) = I$

$$G(s) = G_i(s)\, G_o(s)$$

$\hookrightarrow \in RH_{\infty}$ | si $G_i{}^{\sim} G_i = 1$

$\hookrightarrow \exists G_o^{*}(s) \in RH_{\infty}$ $\quad G_o(s)\, G_o^{*}(s) =$

Factorizarea outer (G_o) reprezinta partea inversabila, de faza minima a sistemului G. Totodata, are toti polii in $\mathbb{C}_-$. Asta inseamna ca G_o sa aiba rang intreg pe linii in $\mathbb{C}_+$, adica are rang normal intreg pe linii si NU ARE POLI IN $C_+ \cup C_0$. Daca matricea este patratica, pot sa iau $G_o^{\#}(s) = G_o^{-1}(s)$

Pe de alta parte, factorizarea inner (G_i) conserva energia si norma infinit (demonstratie mai jos).
De asemenea, acest fel de sistem izoleaza zerourile instabile in $\mathbb{C}_+$. Totodata, pentru matrici patrate,

$G_i^{\sim}(s) = G_i^{-1}(-s)$

De unde am scos asta:

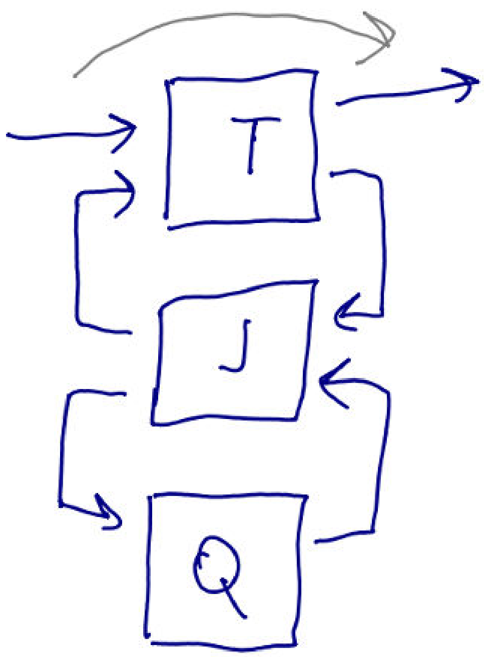
chrome-extension://efaidnbmnnnibpcajpcglclefindmkaj/https://www.math.rug.nl/~arjan/DownloadPublicaties/petersen.pdf

O intrebare normala: de ce aceste proprietati ciudate si random? Hai sa vedem:

Fie un $T \in \mathscr{RH}_\infty$ si factorizarea inner $G_i T \in \mathscr{RH}_\infty$. Sa ne uitam la norma infinit a lui $||G_i T||_\infty = \sup_w \sigma_{\max}(G_i(\mathrm{jw})T(\mathrm{jw})) = \sup_{w\in\mathbb{R}} \lambda_{\max}\big(T^*(\mathrm{jw})G_i^*(\mathrm{jw})G_i(\mathrm{jw})T(\mathrm{jw})\big)$. Vedem ca avem $G_i^*(\mathrm{jw})G_i(\mathrm{jw})$. Daca ne uitam la factorizarea inner, vedem ca avem acelasi format, deci $G_i^*(\mathrm{jw})G_i(\mathrm{jw}) = I$. Inlocuim asta si obtinem $\sup_{w\in\mathbb{R}} \lambda_{\max}(T^*(\mathrm{jw})T(\mathrm{jw})) = ||T||_\infty$. **Deci, factorizarea inner conserva norma infinit a lui T.**

Acum sa ne uitam la norma 2: $||G_i T||_2^2 = \frac{1}{2\pi}\int_{-\infty}^{\infty} \mathrm{tr}(T^* G^* \mathrm{GT}(\mathrm{jw}))\mathrm{dw} = \frac{1}{2\pi}\int_{-\infty}^{\infty} (T^* T(\mathrm{jw}))\mathrm{dw} = ||T||_2^2$. **Deci, factorizarea inner conservas si norma 2 a lui T.**

Sa facem o mica reamintire la Youla:



Practic este un TLFi intre T si TLFi de J si Q: $\mathrm{TLFi}(T, \mathrm{TLFi}(J, Q)) = T_1 + T_2 \mathrm{QT}_3$, $Q \in \mathscr{RH}_\infty$ petru a avea stabillitate din Youla. Fie $\alpha = \{2, \infty\}$. Sa calculam norma α din $\mathrm{TLFi}(T, \mathrm{TLFi}(J, Q)) = T_1 + T_2 \mathrm{QT}_3$:

Mai intai scriem pe T_2 ca factorizare inner-outer

$||T_1 + T_2 QT_3||_\alpha = ||T_1 + T_{2i}T_{2o}QT_3||_\alpha$.

Acum luam pe T_3 si scriem in factorizare **co-inner-outer**: $T_3 = T_{3_{co}}T_{3_{ci}}$, unde

$$\begin{cases} T_{3_{co}} : \exists\, T^{\#}_{3_{co}} \in \mathscr{RH}_\infty\ a.i\ T^{\#}_{3_{co}} T_{3_{co}} = I \\ T_{3_{ci}} : \text{inversa lui } T_{3_{ci}} : \tilde{T}_{3_{ci}} \in \mathscr{RH}_\infty\ a.i\ T_{3_{ci}} \tilde{T}_{3_{ci}} = I \end{cases}$$

↪ co-outer $\exists\, T_{3co}^{*} \in \mathbb{RH}_\infty$ a.i. $T_{3co}^{*} T_{3co} = I$

$T_3 = T_{3co} T_{3ci}$

↳ co inner

$\in \mathbb{RH}_\infty$: $T_{3ci} \tilde{T}_{3ci} = I$

Totusi, $G(s) = G_i(s)G_0(s)$, unde G_I este factorilzare inner $\in \mathscr{RH}_\infty$ si $\tilde{G_i} G_i = I$. G_o este factorizare outer. Aici $\exists G_o^{\#} \in \mathscr{RH}_\infty$ unde $G_o(s)G_o^{\#}(s) = I$. Ce diferente exista? La inner inversa este la stanga, iar la co-inner la dreapta. De asemenea, inner are pe $G_o^{\#}$ la dreapta, iar co-outer are $T^{\#}_{3_{co}}$ la stanga.

Fun fact: Factorizarile inner si co-inner sunt duale, adica $G_i\,\text{inner} \Leftrightarrow G_i^T\ \text{co}-\text{inner}$

Ok,a cum sa reneveim: Aveam $||T_1 + T_{2i}T_{2o}QT_3||_\alpha$ si vrem sa scriem pe $T_3 = T_{3_{co}}T_{3_{ci}}$. Asadar, avem: $||T_1 + T_{2i}T_{2o}QT_{3_{co}}T_{3_{ci}}||_\alpha$. Sa facem notatia $T^{\#}_{2o}\hat{Q}T^{\#}_{3_{co}} = Q$. Asadar, $\hat{Q} = T_{2o}QT_{3_{co}}$ si avem: $||T_1 + T_{2i}\hat{Q}T_{3_{ci}}||_\alpha$. Acum ar fi ceva sa dam factor comun la stanga si dreapta pentru a reduce suma, dar cum?

Facem urmatoarele notatii:

- $T_{2i}^{\perp} \in \mathscr{RH}_\infty$ este completare pana cand matricea $\begin{bmatrix} T_{2i} & T_{2i}^{\perp} \end{bmatrix}$ este matrice patrat si inversabila si respecta factorizarea inner: $\widetilde{\begin{bmatrix} T_{2i} & T_{2i}^{\perp} \end{bmatrix}} \begin{bmatrix} T_{2i} & T_{2i}^{\perp} \end{bmatrix} = I$.

- $T_{3_{\text{ci}}}^{\perp} \in \mathscr{RH}_{\infty}$ este completare pana cand matricea $\begin{bmatrix} T_{3_{\text{ci}}} \\ T_{3_{\text{ci}}}^{\perp} \end{bmatrix}$ este matricea patrat si incersabila si respecta factorizarea co-inner: $\begin{bmatrix} T_{3_{\text{ci}}} \\ T_{3_{\text{ci}}}^{\perp} \end{bmatrix}\begin{bmatrix} T_{3_{\text{ci}}} \\ T_{3_{\text{ci}}}^{\perp} \end{bmatrix}^{\sim} = I$

Reluam pe $||T_1 + T_{2i}\widehat{Q}T_{3_{\text{ci}}}||_{\alpha}$ si adaugam la stanga o factorizare co-inner si la dreapta o factorizare inner:

$||T_l\left[T_l^{\sim}\left(T_1 + T_{2i}\widehat{Q}T_{3_{\text{ci}}}\right)T_r^{\sim}\right]T_r||_{\alpha}$. $||T_l^{\sim}\left(T_1 + T_{2i}\widehat{Q}T_{3_{\text{ci}}}\right)T_r^{\sim}||_{\alpha} = ||T_l^{\sim}T_1T_r^{\sim} + \begin{bmatrix} I \\ 0 \end{bmatrix}\widehat{Q}[I \quad 0]||_{\alpha}$. Fie $T_l^{\sim}T_1T_r^{\sim} = R \rightarrow \text{antistabil (constant } sau \text{ polii in } \mathbb{C}_+)$. De aici rescriem asa: $||\begin{bmatrix} R_{11} + \widehat{Q} & R_{12} \\ R_{21} & R_{22} \end{bmatrix}||_{\alpha}$. Asta este o problema de distanta minima aparent.

Aparent (si din pacate) regulatorul nostru este de ordin 3n (am R de ordin 2n si ordin initial n). Multicel, deci trebuie alese Q-urile pentru a avea regulator de ordin n prin simplificari.

Ok ok, pana sa trec mai departe, simt ca ar trebui sa reexplic ce inseamna sistem minimal pe $\mathbb{C}_0$, sau in englza $\begin{cases} C_0 - \text{controllable} \\ \mathbb{C}_0 - \text{observable} \end{cases}$. De la TSA, partea de spatiul starilor (state-space), noi stim ca un sistem controlabil, folosind Criteriul PBH, respecta asta $\text{rank}[\text{sI} - A \quad B] = n, \forall\, s \in \mathbb{C}$, iar la observabil avem $\text{rank}\begin{bmatrix} \text{sI} - A \\ C \end{bmatrix} = n, \forall\, s \in \mathbb{C}$, unde $s \in \Lambda(A)$.

Acum ca am reamintit asta, minimal pe $\mathbb{C}_0$ inseamna ca $\begin{cases} \text{rank}[\text{sI} - A \quad B] = n, \forall\, s \in \mathbb{C}_0 \\ \text{rank}\begin{bmatrix} \text{sI} - A \\ C \end{bmatrix} = n, \forall\, s \in \mathbb{C}_0 \end{cases}$

Fie sistemul $G(s) = \begin{bmatrix} A & B \\ C & D \end{bmatrix}$, cu $\begin{cases} A \in \mathbb{R}^{n\times n} & B \in \mathbb{R}^{n\times m} \\ C \in \mathbb{R}^{p\times n} & D \in \mathbb{R}^{m\times p} \end{cases}$. Presupunem ca $p \geq m$, adica matricea de stare este ori patrat sau dreptunghi inalt. Mai presupunem ca G are rang normal intreg pe coloane (adica m), nu are 0 pe axa imaginara. Daca avem $\begin{cases} (A, B) \text{ staebilizabil} \\ (C, A)\ \mathbb{C}_0 - \text{observabil} \end{cases}$, atunci:

1. Pentru tripletul Popov $\Sigma = \left(A; B; \begin{bmatrix} C^TC & C^TD \\ D^TC & D^TD \end{bmatrix}\right)$ are o solutie stabilizatoare X si reactie stabilizatoare F

2. Presupunem $H \in \mathbb{R}^{m \times m}$ o **matrice inversabila** astfel incat $H^T H = D^T D$. Facem svd pe D si obtinem: $\operatorname{svd}(D) = U \begin{bmatrix} \Sigma \\ 0 \end{bmatrix} V^T$. Acum facem $\operatorname{svd}(D^T D) = V \Sigma^2 V^T$. H o sa il presupunem choeslky din $D^T D$ *(INFERIOR TRIUNGHIULAR)*. Vom defiini:

$$\begin{cases} G_o(s) = \begin{bmatrix} A & B \\ -\mathrm{HF} & H \end{bmatrix} \\ G_i(s) = \begin{bmatrix} A + \mathrm{BF} & \mathrm{BH}^{-1} \\ C + \mathrm{DF} & \mathrm{DH}^{-1} \end{bmatrix}, A + \mathrm{BF} \text{ este tot aia cu } A - \mathrm{BK}, \text{unde } F = -K \text{ (curs de CO)} \end{cases}$$

Sa vedem cum arata inversa lui $G_o(s)$: $G_o^{-1} = \begin{bmatrix} A - \mathrm{BH}^{-1}(-\mathrm{HF}) & \dots \\ \dots & H^{-1} \end{bmatrix}$. Acele $\dots$ arata ca avem ceva acolo, dar nu are importanta atat de mare

Factorizaru coprime ***NORMALIZATE***

Fie 2 matrici $M, N \in \mathcal{RH}_\infty$. Acestea sunt coprime daca au acelasi numar de coloane si $\exists \widetilde{X}, \widetilde{Y} \in \mathcal{RH}_\infty$ astfel incat sa aiba loc $\begin{bmatrix} \widetilde{Y} & -\widetilde{X} \end{bmatrix} \begin{bmatrix} M \\ N \end{bmatrix} = \widetilde{Y} M - \widetilde{X} N = I$, iar $G = \mathrm{NM}^{-1}$ (Bezout la dreapta)

Fie alte 2 matrici $\widetilde{M}, \widetilde{N} \in \mathcal{RH}_\infty$. Acestea sunt coprime daca au acelasi numar de linii si $\exists X, Y \in \mathcal{RH}_\infty$ astfel incat ca aiba loc $\begin{bmatrix} \widetilde{N} & \widetilde{M} \end{bmatrix} \begin{bmatrix} -X \\ Y \end{bmatrix} = \widetilde{M} Y - \widetilde{N} X = I$, iar $G = \widetilde{M}^{-1} \widetilde{N}$ (Bezout la stanga)

Fie G un sistem aleator. O factorizare coprima la $\begin{cases} \text{dreapta} : G = \mathrm{NM}^{-1} \\ \text{stanga} : G = \widetilde{M}^{-1} \widetilde{N} \end{cases}$ se numeste in plus normalizata daca satisface $\begin{cases} N^{\sim} N + M^{\sim} M = I \\ \widetilde{N} \widetilde{N}^{\sim} + \widetilde{M} \widetilde{M}^{\sim} = I \end{cases}$. Scris in froma matriceala: $\begin{cases} \begin{bmatrix} N \\ M \end{bmatrix}^{\sim} \begin{bmatrix} N \\ M \end{bmatrix} = \begin{bmatrix} N^{\sim} & M^{\sim} \end{bmatrix} \begin{bmatrix} N \\ M \end{bmatrix} = I \\ \begin{bmatrix} \widetilde{N} & \widetilde{N} \end{bmatrix} \begin{bmatrix} \widetilde{N} & \widetilde{M} \end{bmatrix}^{\sim} = I \end{cases}$

Fie matricea $\begin{bmatrix} N \\ M \end{bmatrix} = \begin{bmatrix} A_c & B_c \\ C_c & D_c \end{bmatrix}$ o realizare minimala. Putem rescrie: $\begin{bmatrix} N \\ M \end{bmatrix} = G_i G_o = \begin{bmatrix} G_{i1} \\ G_{i1} \end{bmatrix} G_o$, unde $\begin{cases} G_{i1} = N_m \\ G_{i2} = M_m \end{cases}$.

De aici scoatam ca $\begin{cases} N_m = \mathrm{NG}_o^{-1} \\ M_m = \mathrm{MG}_o^{-1} \end{cases}$

Laborator

```
clear
close all
clc
```

Scriu A,B,C,D

```
A = randn(5)
```

```
A = 5×5
    0.7449   -0.4259    1.5973    0.4431    0.4519
   -0.8282    0.6361    0.1124   -0.1348    1.6484
    0.5745    0.7932   -0.3086   -0.0183   -2.0284
    0.2818   -0.8984    0.4567    0.4608   -0.4493
    1.1393    0.1562   -0.2751    1.3623    0.2360
```

```
B = randn(5,3)
```

```
B = 5×3
   -0.8352    0.3953    0.3319
   -1.2760   -0.8706    2.3652
    0.6170   -0.4977   -0.4822
    0.6127   -0.1067    0.6474
    0.2894   -0.6878   -1.0344
```

```
C = randn(4,5)
```

```
C = 4×5
    1.3396    0.5120   -0.6300    0.5530    0.6521
   -0.9691    0.0114   -0.0469   -1.0765   -0.2789
    0.2087   -0.0440    2.6830    1.0306    0.2452
   -0.6186    2.9491   -1.1467    0.3275    1.4725
```

```
D = randn(4,3)
```

```
D = 4×3
   -2.2751   -0.2963   -0.7258
   -1.6333   -1.4969   -0.8665
    0.4155   -0.9048   -0.4218
   -0.6548   -0.4042   -0.9427
```

Acum le fac in spatiul starilor

```
G = ss(A,B,C,D)
```

```
G =
 
  A = 
             x1        x2        x3        x4        x5
   x1    0.7449   -0.4259     1.597    0.4431    0.4519
   x2   -0.8282    0.6361    0.1124   -0.1348     1.648
   x3    0.5745    0.7932   -0.3086  -0.01833    -2.028
   x4    0.2818   -0.8984    0.4567    0.4608   -0.4493
   x5     1.139    0.1562   -0.2751     1.362     0.236
 
  B = 
            u1       u2       u3
   x1  -0.8352   0.3953   0.3319
   x2   -1.276  -0.8706    2.365
   x3    0.617  -0.4977  -0.4822
```

```
   x4   0.6127  -0.1067   0.6474
   x5   0.2894  -0.6878   -1.034

  C =
             x1        x2        x3        x4        x5
   y1      1.34     0.512     -0.63     0.553    0.6521
   y2   -0.9691   0.01135  -0.04688    -1.076   -0.2789
   y3    0.2087  -0.04399     2.683     1.031    0.2452
   y4   -0.6186     2.949    -1.147    0.3275     1.473

  D =
            u1       u2       u3
   y1   -2.275  -0.2963  -0.7258
   y2   -1.633   -1.497  -0.8665
   y3   0.4155  -0.9048  -0.4218
   y4  -0.6548  -0.4042  -0.9427

Continuous-time state-space model.
Model Properties
```

Valori proprii ale lui A

```
eig(A)
```

```
ans = 5×1 complex
  -2.2542 + 0.0000i
   1.4457 + 1.3864i
   1.4457 - 1.3864i
   0.5659 + 0.2283i
   0.5659 - 0.2283i
```

O mica reamintire:

Fie $G(s) = \begin{bmatrix} A & B \\ C & D \end{bmatrix}$, cu $\begin{cases} (A, B) \text{ stabilizabil} \\ (C, A) \text{ detectabil} \end{cases}$. In acest sens, fie F si L astfel incat $\begin{cases} \Lambda(A + \mathrm{BF}) \in \mathbb{C}_- \\ \Lambda(A + \mathrm{LC}) \in \mathbb{C}_- \end{cases}$. Atunci

$$\begin{cases} \begin{bmatrix} M & X \\ N & Y \end{bmatrix} = \begin{bmatrix} A + \mathrm{BF} & B & -L \\ F & I & 0 \\ C + \mathrm{DF} & 0 & I \end{bmatrix} \\ \begin{bmatrix} \widetilde{Y} & -\widetilde{X} \\ -\widetilde{N} & \widetilde{M} \end{bmatrix} = \begin{bmatrix} A + \mathrm{LC} & -(B + \mathrm{LD}) & L \\ F & I & 0 \\ C & -D & I \end{bmatrix} \end{cases}$$

$$\left[\begin{array}{cc} \mathbf{M} & \mathbf{X} \\ \mathbf{N} & \mathbf{Y} \end{array}\right] = \left[\begin{array}{c|cc} A+BF & B & -L \\ \hline F & I & 0 \\ C+DF & D & I \end{array}\right]$$

$$\left[\begin{array}{cc} \widetilde{\mathbf{Y}} & -\widetilde{\mathbf{X}} \\ -\widetilde{\mathbf{N}} & \widetilde{\mathbf{M}} \end{array}\right] = \left[\begin{array}{c|cc} A+LC & -(B+LD) & L \\ \hline F & I & 0 \\ C & -D & I \end{array}\right]$$

reprezintă o factorizare dublu coprimă (56), (57) a lui **G**. . Atunci J va avea forma

$$J = \begin{bmatrix} A + \mathrm{BF} + \mathrm{LC} + \mathrm{LDF} & -L & B + \mathrm{LD} \\ F & 0 & I \\ -(C + \mathrm{DF}) & I & -D \end{bmatrix}$$

$G = \mathrm{NM}^{-1}, (N, M)$ factorizarte coprima la dreapta pe $\mathscr{RH}_\infty$

```
F = -place(A,B,[-4 -5 -3 -6 -9])
```

```
F = 3×5
  -21.1267    5.4842  -13.8306  -18.0321   -0.2735
  -23.6707    3.1601  -23.8865   -5.2489   13.8928
  -13.6601   -0.6008  -16.8387   -4.4535    9.4937
```

```
L = -place(A',C',[-4 -5 -3 -6 -9])'
```

```
L = 5×4
   -5.9478   -3.8148   -1.7910    1.7347
   -1.2373   -2.6033   -1.4819   -1.4685
   -1.2678   -3.5934   -3.5291    1.2442
    4.5726    7.1108    0.8076   -0.5111
   -1.8461    0.9198    1.2615   -1.7543
```

```
M = ss(A + B*F, B, F, eye(3))
```

```
M =

  A = 
           x1      x2      x3      x4      x5
   x1   4.498  -3.956  -1.883   11.95   9.323
   x2   14.43  -10.53  -1.273   16.91   12.36
   x3   5.907   2.894   11.17  -6.385  -13.69
   x4  -18.98   1.736  -16.37  -12.91   4.048
   x5   25.44  0.1911   29.57   4.361  -19.22

  B = 
            u1       u2       u3
   x1  -0.8352   0.3953   0.3319
   x2   -1.276  -0.8706    2.365
   x3    0.617  -0.4977  -0.4822
   x4   0.6127  -0.1067   0.6474
   x5   0.2894  -0.6878   -1.034

  C = 
            x1       x2       x3       x4       x5
   y1   -21.13    5.484   -13.83   -18.03  -0.2735
```

```
  y2   -23.67     3.16   -23.89   -5.249    13.89
  y3   -13.66  -0.6008   -16.84   -4.453    9.494

 D =
      u1  u2  u3
  y1   1   0   0
  y2   0   1   0
  y3   0   0   1

Continuous-time state-space model.
Model Properties
```

```
N = ss(A + B*F, B, C + D*F, D)
```

```
N =

 A =
          x1      x2      x3      x4      x5
  x1   4.498  -3.956  -1.883   11.95   9.323
  x2   14.43  -10.53  -1.273   16.91   12.36
  x3   5.907   2.894   11.17  -6.385  -13.69
  x4  -18.98   1.736  -16.37  -12.91   4.048
  x5   25.44  0.1911   29.57   4.361  -19.22

 B =
           u1       u2       u3
  x1  -0.8352   0.3953   0.3319
  x2   -1.276  -0.8706    2.365
  x3    0.617  -0.4977  -0.4822
  x4   0.6127  -0.1067   0.6474
  x5   0.2894  -0.6878   -1.034

 C =
           x1       x2       x3       x4       x5
  y1    66.33   -12.47    50.14    46.37   -9.733
  y2    80.81   -13.16    72.89    40.09   -28.85
  y3    18.61  -0.3714    25.65    0.167   -16.44
  y4    35.66   -1.353    33.44    18.45   -12.91

 D =
           u1       u2       u3
  y1   -2.275  -0.2963  -0.7258
  y2   -1.633   -1.497  -0.8665
  y3   0.4155  -0.9048  -0.4218
  y4  -0.6548  -0.4042  -0.9427

Continuous-time state-space model.
Model Properties
```

Vrem sa facem asta: $\begin{bmatrix} N \\ M \end{bmatrix} = G_i G_o = \begin{bmatrix} G_{\mathrm{i1}} \\ G_{\mathrm{i1}} \end{bmatrix} G_o$, cu $\begin{cases} G_{\mathrm{i1}} = N_m \\ G_{\mathrm{i2}} = M_m \end{cases}$.

```
col_rf = [M;
          N]
```

```
col_rf =

 A =
           x1       x2       x3       x4       x5       x6       x7       x8       x9      x10
  x1    4.498   -3.956   -1.883    11.95    9.323        0        0        0        0        0
  x2    14.43   -10.53   -1.273    16.91    12.36        0        0        0        0        0
```

```
  x3    5.907   2.894   11.17  -6.385  -13.69       0       0       0       0       0
  x4   -18.98   1.736  -16.37  -12.91   4.048       0       0       0       0       0
  x5    25.44  0.1911   29.57   4.361  -19.22       0       0       0       0       0
  x6        0       0       0       0       0   4.498  -3.956  -1.883   11.95   9.323
  x7        0       0       0       0       0   14.43  -10.53  -1.273   16.91   12.36
  x8        0       0       0       0       0   5.907   2.894   11.17  -6.385  -13.69
  x9        0       0       0       0       0  -18.98   1.736  -16.37  -12.91   4.048
  x10       0       0       0       0       0   25.44  0.1911   29.57   4.361  -19.22

 B =
            u1       u2       u3
  x1   -0.8352   0.3953   0.3319
  x2    -1.276  -0.8706    2.365
  x3     0.617  -0.4977  -0.4822
  x4    0.6127  -0.1067   0.6474
  x5    0.2894  -0.6878   -1.034
  x6   -0.8352   0.3953   0.3319
  x7    -1.276  -0.8706    2.365
  x8     0.617  -0.4977  -0.4822
  x9    0.6127  -0.1067   0.6474
  x10   0.2894  -0.6878   -1.034

 C =
           x1       x2       x3       x4       x5       x6       x7       x8       x9      x10
  y1   -21.13    5.484   -13.83   -18.03  -0.2735        0        0        0        0        0
  y2   -23.67     3.16   -23.89   -5.249    13.89        0        0        0        0        0
  y3   -13.66  -0.6008   -16.84   -4.453    9.494        0        0        0        0        0
  y4        0        0        0        0        0    66.33   -12.47    50.14    46.37   -9.733
  y5        0        0        0        0        0    80.81   -13.16    72.89    40.09   -28.85
  y6        0        0        0        0        0    18.61  -0.3714    25.65    0.167   -16.44
  y7        0        0        0        0        0    35.66   -1.353    33.44    18.45   -12.91

 D =
           u1       u2       u3
  y1        1        0        0
  y2        0        1        0
  y3        0        0        1
  y4   -2.275  -0.2963  -0.7258
  y5   -1.633   -1.497  -0.8665
  y6   0.4155  -0.9048  -0.4218
  y7  -0.6548  -0.4042  -0.9427

Continuous-time state-space model.
Model Properties
```

Cum am zis trebuie sa facem realizare minimala

Daca avem $\begin{cases} (A, B) \text{ staebilizabil} \\ (C, A)\ \mathbb{C}_0 - \text{observabil} \end{cases}$, atunci:

1. Pentru tripletul Popov $\Sigma = \left(A; B; \begin{bmatrix} C^T C & C^T D \\ D^T C & D^T D \end{bmatrix} \right)$ are o solutie stabilizatoare X si reactie stabilizatoare F
2. Presupunem $H \in \mathbb{R}^{m \times m}$ o **matrice inversabila** astfel incat $H^T H = D^T D$. Facem svd pe D si obtinem: $\text{svd}(D) = U \begin{bmatrix} \Sigma \\ 0 \end{bmatrix} V^T$. Acum facem $\text{svd}(D^T D) = V \Sigma^2 V^T$. H o

sa il presupunem choeslky din $D^T D$ *(INFERIOR TRIUNGHIULAR)*. Vom defiini:

$$\begin{cases} G_o(s) = \begin{bmatrix} A & B \\ -\mathrm{HF} & H \end{bmatrix} \\ G_i(s) = \begin{bmatrix} A+\mathrm{BF} & \mathrm{BH}^{-1} \\ C+\mathrm{DF} & \mathrm{DH}^{-1} \end{bmatrix}, A+\mathrm{BF} \text{ este tot aia cu } A-\mathrm{BK}, \text{unde } F=-K \text{ (curs de CO)} \end{cases}$$

```
col_rf = ss(col_rf, 'min')
```

```
col_rf =

  A = 
             x1        x2        x3        x4        x5
   x1     6.111      11.4    -4.071  -0.03133  -0.04142
   x2    -7.111    -16.09    -1.025  0.004664    0.9674
   x3     6.002     1.797    -11.51  0.002142     1.166
   x4    -26.24     6.287     42.33    -4.657    -9.093
   x5    -10.07    -25.19    -10.93   -0.1199   -0.8589

  B = 
             u1       u2       u3
   x1   -0.9211   0.6844  -0.2206
   x2     1.382  -0.4071  -0.8343
   x3   -0.2725   0.4424   -1.166
   x4    0.3012      1.4   0.5303
   x5     1.846   0.7097   -3.531

  C = 
            x1       x2       x3       x4       x5
   y1   -13.65   -8.506    15.03   -2.101   -2.764
   y2   -12.87    5.309    21.12   -2.426   -5.908
   y3   -6.711    3.389    15.04   -1.034   -2.519
   y4    40.71    15.73   -51.28    6.123    9.199
   y5     46.7    2.381   -68.79    7.818    15.77
   y6    8.729   -9.247   -20.97    1.122    5.887
   y7    19.97   0.6482   -32.02    2.248    4.462

  D = 
             u1       u2       u3
   y1         1        0        0
   y2         0        1        0
   y3         0        0        1
   y4    -2.275  -0.2963  -0.7258
   y5    -1.633   -1.497  -0.8665
   y6    0.4155  -0.9048  -0.4218
   y7   -0.6548  -0.4042  -0.9427

Continuous-time state-space model.
Model Properties
```

Factorizare coprima normalizata, M si N normalizati (facem inner-outer)

Facem triplet Popov $\Sigma_{\mathrm{io}} = \left(A_r; B_r; \begin{bmatrix} C_r^T C_r & C_r^T D_r \\ D_r^T C_r & D_r^T D_r \end{bmatrix} \right)$

```
col_rf.D
```

```
ans = 7×3
```

```
    1.0000         0         0
         0    1.0000         0
         0         0    1.0000
   -2.2751   -0.2963   -0.7258
   -1.6333   -1.4969   -0.8665
    0.4155   -0.9048   -0.4218
   -0.6548   -0.4042   -0.9427
```

```
rank(col_rf.D) % Aici verificam ca avem rang intreg pe coloane, adica nu avem 0-uri la infinit
```

```
ans =
3
```

Am vazut ca nu pica rangul, deci nu avem 0-uri infinite

```
tzero(col_rf)
```

```
ans =

  0×1 empty double column vector
```

ACum vedem $\mathrm{EMAR}(\Sigma)$

$$\Sigma_{\mathrm{io}} := (A;B;P) = (A;B;Q,L,R), \begin{bmatrix} Q & L \\ L^T & R \end{bmatrix} = P = P^T$$

$$Q = C^T C\,,\ L = C^T D,\ R = D^T D$$

```
[X_r,F_r, spectru_r] = icare(col_rf.A, col_rf.B, col_rf.C'*col_rf.C,
col_rf.D'*col_rf.D, col_rf.C'*col_rf.D)
```

```
X_r = 5×5
   35.6419   13.2751  -46.3178    8.4576    8.7392
   13.2751   17.8139   -3.9643   -1.4188   -2.5841
  -46.3178   -3.9643   77.5241  -17.6629  -17.8205
    8.4576   -1.4188  -17.6629    5.8903    5.0664
    8.7392   -2.5841  -17.8205    5.0664    5.5668
F_r = 3×5
  -12.9492   -6.0620   15.7767   -2.3540   -2.7484
   -8.9161    4.7420   14.9968    0.2126   -3.6605
   -7.4259   -0.3336   11.4061   -0.5763   -2.2571
spectru_r = 5×1 complex
  -0.2763 + 0.0000i
  -1.8225 + 1.0799i
  -1.8225 - 1.0799i
  -3.0219 + 0.0000i
  -6.1114 + 0.0000i
```

```
F_r = -F_r; % Facem reactie pozitiva, de-aia
help chol
```

```
 chol - Cholesky factorization

    Syntax
      Full or Sparse Data
        R = chol(A)
        R = chol(A,triangle)
        [R,flag] = chol(___)
```

```
  Sparse Data
    [R,flag,P] = chol(S)
    [R,flag,P] = chol(___,outputForm)

Input Arguments
  A - Input matrix
    matrix
  S - Sparse input matrix
    sparse matrix
  triangle - Triangular factor of input matrix
    'upper' (default) | 'lower'
  outputForm - Shape of permutation output
    'matrix' (default) | 'vector'

Output Arguments
  R - Cholesky factor
    matrix
  flag - Symmetric positive definite flag
    scalar
  P - Permutation for sparse matrices
    matrix | vector

Examples
  Solve Linear System with Symmetric Positive Definite Matrix
  Cholesky Factorization of Matrix
  Suppress Errors for Nonsymmetric Positive Definite Matrices
  Cholesky Factor of Sparse Matrix
  Reorder Sparse Matrix with Permutation Vector

See also ichol, cholupdate, ldl, qr

Introduced in MATLAB before R2006a
Documentation for chol
Other uses of chol
```

Putin matlab time: Matlab la cholesky face L^TL; Noua matlab ne da matrice superior triunghiulara by default. Putem face chol(matrice, 'lower') ca sa ne dea inferior triunghiualra, dupa transpunem.

Cholesky normal: LL^T

```
H = chol(col_rf.D'*col_rf.D, 'lower')'
```

```
H = 3×3
    3.0733    0.9787    1.1416
         0    1.8311    0.6322
         0         0    1.2811
```

$$\mathbf{G}_o(s) \quad := \quad \left[\begin{array}{c|c} A & B \\ \hline -HF & H \end{array}\right],$$

```
G_o = ss(col_rf.A, col_rf.B, -H*F_r, H)
```

```
G_o =

  A =
             x1         x2         x3         x4         x5
```

```
   x1     6.111      11.4    -4.071  -0.03133  -0.04142
   x2    -7.111    -16.09    -1.025  0.004664    0.9674
   x3     6.002     1.797    -11.51  0.002142     1.166
   x4    -26.24     6.287     42.33    -4.657    -9.093
   x5    -10.07    -25.19    -10.93   -0.1199   -0.8589

  B = 
            u1       u2       u3
   x1  -0.9211   0.6844  -0.2206
   x2    1.382  -0.4071  -0.8343
   x3  -0.2725   0.4424   -1.166
   x4   0.3012      1.4   0.5303
   x5    1.846   0.7097   -3.531

  C = 
            x1       x2       x3       x4       x5
   y1      -57   -14.37    76.18   -7.685   -14.61
   y2   -21.02    8.472    34.67  0.02491   -8.129
   y3   -9.513  -0.4273    14.61  -0.7384   -2.892

  D = 
           u1      u2      u3
   y1   3.073  0.9787   1.142
   y2       0   1.831  0.6322
   y3       0       0   1.281

Continuous-time state-space model.
Model Properties
```

Acum pregatec partile pentru $G_{\mathrm{i}1}$ si $G_{\mathrm{i}2}$, mai specific $C_{\mathrm{r}1}, C_{\mathrm{r}2}, D_{\mathrm{r}1}, D_{\mathrm{r}2}$

Ideea de baza: stim ca $G_i = \begin{bmatrix} A+\mathrm{BF} & \mathrm{BH}^{-1} \\ C+\mathrm{DF} & \mathrm{DH}^{-1} \end{bmatrix}$. Insa noi am scris $G_i = \begin{bmatrix} G_{\mathrm{i}1} \\ G_{12} \end{bmatrix}$. A si B din G_i nu ne afecteaza, ci C si D. Asadar, rescriem: $\begin{bmatrix} G_{\mathrm{i}1} \\ G_{12} \end{bmatrix} = \begin{bmatrix} A+\mathrm{BF} & \mathrm{BH}^{-1} \\ C_1+D_1F & D_1H^{-1} \\ C_2+D_2F & D_2H^{-1} \end{bmatrix}$

```
C_r1 = col_rf.C(1:3,:)
```

```
C_r1 = 3×5
  -13.6479   -8.5057   15.0267   -2.1005   -2.7642
  -12.8738    5.3090   21.1186   -2.4258   -5.9081
   -6.7105    3.3888   15.0434   -1.0344   -2.5186
```

```
C_r2 = col_rf.C(4:7,:)
```

```
C_r2 = 4×5
   40.7142   15.7252  -51.2819    6.1229    9.1992
   46.6991    2.3811  -68.7856    7.8180   15.7697
    8.7286   -9.2470  -20.9743    1.1215    5.8874
   19.9658    0.6482  -32.0207    2.2480    4.4624
```

```
D_r1 = col_rf.D(1:3,:)
```

```
D_r1 = 3×3
     1     0     0
     0     1     0
     0     0     1
```

```
D_r2 = col_rf.D(4:7,:)
```

```
D_r2 = 4×3
   -2.2751   -0.2963   -0.7258
   -1.6333   -1.4969   -0.8665
    0.4155   -0.9048   -0.4218
   -0.6548   -0.4042   -0.9427
```

$$\mathbf{G}_i(s) \quad := \quad \left[\begin{array}{c|c} A + BF & BH^{-1} \\ \hline C + DF & DH^{-1} \end{array}\right]$$

```
G_i1 = ss(col_rf.A + col_rf.B * F_r, col_rf.B * eye(size(H)) / H, C_r1 + D_r1 *
F_r, D_r1 * eye(size(H)) / H)
```

```
G_i1 =

  A = 
            x1       x2       x3       x4       x5
   x1   -1.352    2.496    2.713   -2.472  -0.5657
   x2   0.9589   -6.058   -7.207    2.864    1.392
   x3   -2.238   -2.342  -0.5477   -1.405   -0.594
   x4   -5.922    1.652    10.54   -3.939   -1.944
   x5   -6.053   -18.54   -10.42     2.04   -1.157

  B = 
             u1         u2         u3
   x1   -0.2997     0.5339    -0.1686
   x2    0.4497    -0.4627    -0.8237
   x3  -0.08867      0.289    -0.9735
   x4     0.098      0.712  -0.02475
   x5    0.6007    0.06656     -3.324

  C = 
             x1         x2         x3         x4         x5
   y1   -0.6987     -2.444      -0.75     0.2535   -0.01581
   y2    -3.958     0.5669      6.122     -2.638     -2.248
   y3    0.7153      3.722      3.637    -0.4581    -0.2616

  D = 
            u1       u2       u3
   y1   0.3254  -0.1739  -0.2041
   y2        0   0.5461  -0.2695
   y3        0        0   0.7806

Continuous-time state-space model.
Model Properties
```

```
size(M) % Am atasat dimensiunile pentru a vedea de ce am luat pentru G_i1 si G_i2 C
si D asa
```

```
State-space model with 3 outputs, 3 inputs, and 5 states.
```

```
size(G_i1)
```

```
State-space model with 3 outputs, 3 inputs, and 5 states.
```

$$\mathbf{G}_i(s) \quad := \quad \left[\begin{array}{c|c} A+BF & BH^{-1} \\ \hline C+DF & DH^{-1} \end{array}\right]$$

```
G_i2 = ss(col_rf.A + col_rf.B * F_r, col_rf.B * eye(size(H)) / H, C_r2 + D_r2 *
F_r, D_r2 * eye(size(H)) / H)
```

```
G_i2 =

  A =
            x1       x2       x3       x4       x5
   x1   -1.352    2.496    2.713   -2.472  -0.5657
   x2   0.9589   -6.058   -7.207    2.864    1.392
   x3   -2.238   -2.342  -0.5477   -1.405   -0.594
   x4   -5.922    1.652    10.54   -3.939   -1.944
   x5   -6.053   -18.54   -10.42     2.04   -1.157

  B =
             u1        u2        u3
   x1   -0.2997    0.5339   -0.1686
   x2    0.4497   -0.4627   -0.8237
   x3  -0.08867     0.289   -0.9735
   x4     0.098     0.712  -0.02475
   x5    0.6007   0.06656    -3.324

  C =
            x1       x2       x3       x4       x5
   y1    3.222    3.097   -2.665   0.4119   0.2233
   y2    5.768  -0.7105   -10.69    3.792    3.846
   y3    2.908   -2.578   -9.148    2.049    2.765
   y4   0.8833   -1.719   -4.877   0.2493  -0.9443

  D =
             u1        u2        u3
   y1   -0.7403    0.2338  -0.02226
   y2   -0.5314   -0.5335   0.06046
   y3    0.1352   -0.5664   -0.1702
   y4   -0.2131   -0.1069   -0.4932

Continuous-time state-space model.
Model Properties
```

```
size(N)
```

```
State-space model with 4 outputs, 3 inputs, and 5 states.
```

```
size(G_i2)
```

```
State-space model with 4 outputs, 3 inputs, and 5 states.
```

Verificam daca este norma la dreapta (nu facem $G = \mathrm{NM}^{-1}$, ci $\mathrm{GM} = N$)

```
norm(G_i2 - G * G_i1, 'inf')
```

```
ans =
7.1726e-12
```

Acum vedem daca G_o este outer

Aici vad ca G_o stabil

```
eig(G_o)
```

```
ans = 5×1
   -9.0000
   -3.0000
   -6.0000
   -5.0000
   -4.0000
```

Aici va ca G_o^{-1} este stabila

```
eig(eye(size(G_o)) / G_o)
```

```
ans = 5×1 complex
  -6.1114 + 0.0000i
  -0.2763 + 0.0000i
  -1.8225 + 1.0799i
  -1.8225 - 1.0799i
  -3.0219 + 0.0000i
```

Un alt mod interesant pentru a afla valorile proprii

```
eig(G_o.A - G_o.B * (G_o.D\G_o.C))
```

```
ans = 5×1 complex
  -6.1114 + 0.0000i
  -0.2763 + 0.0000i
  -1.8225 + 1.0799i
  -1.8225 - 1.0799i
  -3.0219 + 0.0000i
```

Acum verific ca $\begin{bmatrix} \widehat{M} \\ \widehat{N} \end{bmatrix} = \begin{bmatrix} M \\ N \end{bmatrix} G_o = \begin{bmatrix} G_{\mathrm{i}1} \\ G_{\mathrm{i}2} \end{bmatrix} G_o$

```
norm(col_rf-[G_i1;G_i2]*G_o, 'inf')
```

```
ans =
1.8842e-14
```

Merge, acum vedem daca este si normalizata, adica $\widehat{M}^{\sim} \widehat{M} + \widehat{N}^{\sim} \widehat{N} = I$. Tot asa vedem ca am facut factorizarea inner ok

```
norm(G_i1'*G_i1 + G_i2'*G_i2 - eye(3), 'inf')
```

```
ans =
2.5702e-12
```